

Lab403: Multi-App Integration (Telegram + Airtable)

Marco Martins

TABLE OF CONTENTS

Lab403: Multi-App Integration (Telegram + Airtable)	1
Scenario	2
Tools Used	2
Workflow Overview	2
Step 1: Airtable Setup	3
Step 2: Telegram Bot Setup	4
Step 3: n8n Workflow Configuration	4
Telegram Trigger Node	4
Set Node — Data Transformation	4
Airtable Node	5
Data Transformation Explained	5
Testing & Results	6
How This Workflow Could Be Extended	6
Success Criteria Checklist	7

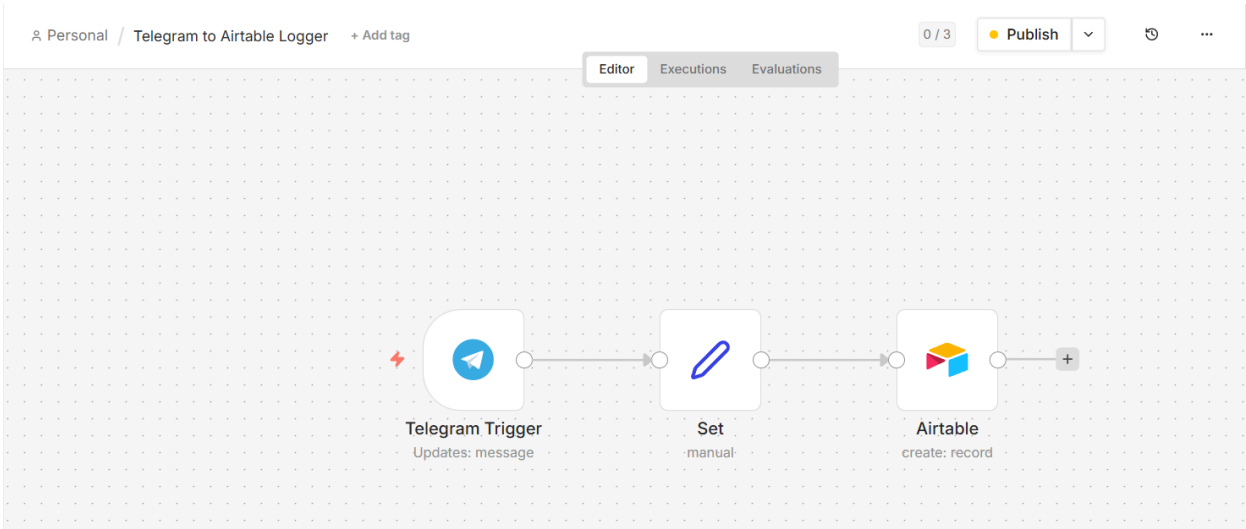
Scenario

As an AI consultant, I automated the logging of client questions received via Telegram into an Airtable tracker. This workflow eliminates manual data entry, enabling better follow-up, response time tracking, and question analysis.

Tools Used

Original Lab	My Implementation
Discord	Telegram
Google Sheets	Airtable
n8n	n8n

Workflow Overview



The workflow consists of 3 nodes:

Telegram Trigger → Set (Transform Data) → Airtable (Create Record)

- **Telegram Trigger** — listens for incoming messages sent to a custom bot (`@client_questions_logger_bot`) and fires in real time via webhook
 - **Set Node** — transforms and maps the raw Telegram payload into a clean structure for Airtable
 - **Airtable Node** — creates a new record in the "Client Questions Tracker" base with the transformed data
-

Step 1: Airtable Setup

Created a new Airtable base named "Client Questions Tracker" with the following fields:

Field	Type
Timestamp	Date & Time
Client Name	Single line text
Chat ID	Single line text
Question	Long text
Status	Single select

Generated a **Personal Access Token** with the following scopes:

- `data.records:read`
- `data.records:write`
- `schema.bases:read`

Step 2: Telegram Bot Setup

Created a new Telegram bot via @BotFather:

- Bot name: `Client Questions Logger`
 - Username: `@client_questions_logger_bot`
 - Obtained the bot API token for use in n8n credentials
-

Step 3: n8n Workflow Configuration

Telegram Trigger Node

- Configured with the bot API token as a credential
- Set to listen for `message` updates
- n8n automatically registered the webhook with Telegram's API upon workflow activation — no manual webhook URL setup required

Set Node — Data Transformation

Telegram delivers a deeply nested JSON payload. The Set node extracted and renamed the relevant fields into a flat structure compatible with Airtable:

Airtable Field	Expression	Source
Timestamp	<code>={{ \$now }}</code>	Generated at runtime
Client Name	<code>={{ \$json.message.from.first_name }}</code>	Telegram sender name
Chat ID	<code>={{ \$json.message.chat.id }}</code>	Telegram chat identifier

Airtable Field	Expression	Source
Question	={{ \$json.message.text }}	Message content
Status	New (fixed value)	Default value

Airtable Node

- Operation: Create Record
 - Base ID: `appiPyZYFJR2NrEZ6`
 - Table ID: `tbls1SQMrx1Kb4Bx1`
 - Mapping mode: Manual — each field mapped to the Set node's output
 - Authenticated via Personal Access Token
-

Data Transformation Explained

Telegram's raw webhook payload is deeply nested:

```
{
  "message": {
    "from": { "first_name": "John" },
    "chat": { "id": 123456789 },
    "text": "How do I implement RAG for my documents?"
  }
}
```

The Set node flattened and renamed this into:

```
{
  "Timestamp": "2025-01-01T10:00:00Z",
```

```

"Client Name": "John",

"Chat ID": "123456789",

"Question": "How do I implement RAG for my documents?",

"Status": "New"

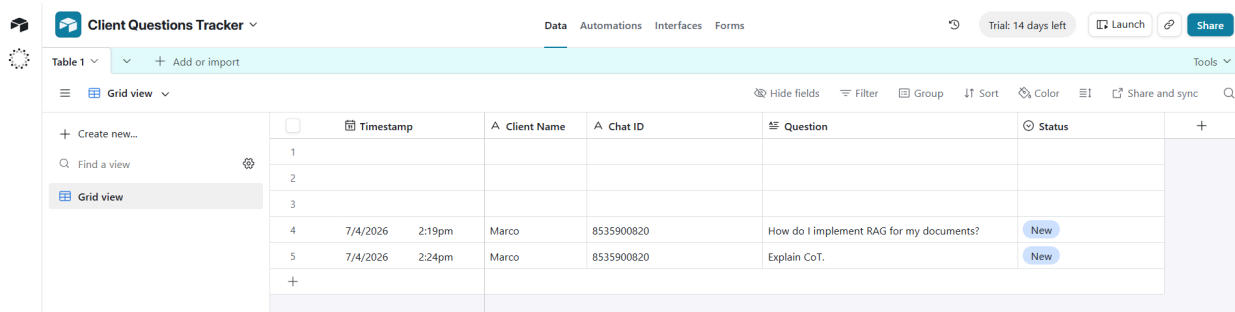
}

```

This transformation is critical because Airtable expects a flat key-value structure, while Telegram delivers nested objects. The Set node acts as the **translation layer** between the two systems.

Testing & Results

- Activated the workflow in n8n (which auto-registered the Telegram webhook)
- Sent test messages to `@client_questions_logger_bot` via Telegram Web
- Verified execution history in n8n showed successful runs
- Confirmed new records appeared in Airtable with all 5 fields correctly populated



	Timestamp	A Client Name	A Chat ID	Question	Status
1					
2					
3					
4	7/4/2026 2:19pm	Marco	8535900820	How do I implement RAG for my documents?	New
5	7/4/2026 2:24pm	Marco	8535900820	Explain CoT.	New

How This Workflow Could Be Extended

- **Keyword filtering** — add an IF node to only log messages containing words like "help", "how", or "question"
- **AI categorization** — route the question through an AI node to auto-tag it as Technical, Billing, or General before logging
- **Auto-reply** — add a Telegram node after the trigger to send an acknowledgement back to the client ("Thanks! I'll get back to you shortly.")

- **Follow-up reminders** — connect to a scheduling tool to create a reminder if a question hasn't been answered within 24 hours
 - **Duplicate detection** — check Airtable before creating a record to avoid logging the same question twice
-

Success Criteria Checklist

- ☒ Successfully created Telegram bot and connected to n8n
- ☒ Configured Airtable integration with correct scopes
- ☒ Transformed nested Telegram data to flat Airtable format
- ☒ Workflow automatically logs messages in real time
- ☒ All 5 columns correctly populated in Airtable
- ☒ Workflow executes without errors